🧪

Ananya Website Evidence : Browser QA and Screenshot Notes

Purpose

This page is the working evidence checklist for Ananya's website-analysis contribution to the single research paper, **Cryptography Through Abstract Algebra — An Interactive Study of Three Transformation Systems**. It is not a separate paper and must not be used to claim tests that have not actually been run.

Scope of Ananya's dossier

Document the actual seven website pages and explain how each supports the mathematical investigation of Caesar shifts, invertible affine transformations modulo 26, and fixed-block position permutations in S4. Collect genuine browser evidence, screenshots with captions, usability observations, and limitations. Do not write a complete paper or modify application code.

Current project evidence

- Repository: final-algebra-lab
- Local project path: D:DownloadsAbstract Algebra
- Branch and known commit: `main`, commit `13d2a3c` (Improve team profiles and cryptographic lab)
- Mathematical source of truth: Caesar C26, affine maps E_(a,b) modulo 26 with gcd(a,26)=1, and S4 position permutations using the documented one-based convention.

- Verified by command execution: `npm run verify` passed; `npm run lint` passed after removing the genuinely unused `caesarInvResult` declaration.
- Not browser-verified: the Vite development server could not start because Windows denied access to Vite/TypeScript cache paths under `node_modules` (EPERM). No screenshot should currently be labelled as browser-tested.
- Build status: the build remains blocked by the same cache-write permission problem. Preserve this as a limitation until repaired and rerun.

Seven-page observation checklist

For each page record: URL/route, visible purpose, mathematical concept, controls, input/output behaviour, source path if identifiable, screenshot filename, exact test date, browser/viewport, and status.

1. Overview — project question, three systems, educational/security disclaimer.
2. Foundations — group concepts: operation, identity, inverse, composition, order, cyclicity, commutativity, subgroup/Lagrange context.
3. The Lab — transformation selection, input validation, encryption/decryption, and round-trip demonstration.
4. Group Explorer — operation tables or structural exploration exposed by the implementation.
5. Experiments — worked transformation comparisons and any brute-force or frequency demonstrations actually present.
6. The Team — profile selection must be click/keyboard driven; hover may only produce visual border glow; GitHub CTA must be checked.
7. Research Log — project evidence, verification notes, limitations, and links actually visible in the UI.

If the current source has a different route or page title, record the discrepancy instead of silently normalising it.

Browser test matrix

Mark each item **Browser verified**, **Source-only**, or **Blocked**, with evidence.

- Navigation reaches each of the seven pages without a console-visible failure.
- Caesar: valid key, encryption, decryption, and round-trip recovery.
- Affine: valid multiplier/shift, invalid non-invertible multiplier rejection, encryption, decryption, and round-trip.
- S4: valid permutation, inverse/decryption, complete four-letter block behaviour, and incomplete final block policy.
- Identity/inverse/composition explanations match the approved mathematics.
- Team page: clicking one profile opens it; clicking it again closes it; clicking another switches the single open profile.
- Team page: hovering changes only visual styling and never reveals content.
- Team page: keyboard Enter/Space activates profile controls; `aria-expanded` and `aria-controls` reflect the open profile.
- Team page: narrow viewport remains readable and controls remain usable.
- Image fallback appears when an image is unavailable, if this is intentionally testable.
- GitHub CTA opens the intended repository URL.
- No claim is complete without a screenshot, observation note, or reproducible test record.

Screenshot and caption template

For every screenshot save: `ANANYA_<page>_<case>_<date>.<format>`.

Caption format: “<Page> — <state/action>; viewport <width × height>; observed result; date; browser.”

Keep screenshots free of private data. Record failed or blocked cases as evidence too, including the EPERM message and the command used.

Evidence boundaries

- **Established mathematics:** the approved mathematical specification and proofs.

- **Verified by code:** output from `npm run verify`, `npm run lint`, and any later reproducible tests.
- **Present in source:** behaviour visible in React/TypeScript/CSS but not exercised in a browser.
- **Browser verified:** only behaviour directly observed after starting the app.
- **Unresolved:** anything blocked by permissions, missing route, unclear source behaviour, or unavailable screenshot.

Dependencies and next actions

1. Project lead/Dotto must repair or provide a user-writable Vite/TypeScript cache location without deleting unrelated work.
2. Start the existing Vite server and record the exact URL, browser, viewport, and date.
3. Execute the matrix above and capture only genuine screenshots.
4. Compare observed labels and mathematical explanations with the approved specification.
5. Hand the dossier, screenshot index, and unresolved-issues list to Pavani for paper integration.
6. Keep GitHub as the source of code/test evidence and Notion as the source of planning, blockers, and progress.

Deadline note

The workspace contains a 31 October 2026 project date, while the latest execution instruction used 24 October 2026. Treat the deadline as unresolved until the coordinator confirms the official submission date; do not silently choose one.

Completion rule

This page is complete only when every claimed browser result has reproducible evidence, blocked items are explicitly marked, and no source-only observation is presented as a live website result.

Browser verification update — 18 October 2026

The local Vite site was opened at http://127.0.0.1:5173/ in the connected browser.

Browser-verified observations

- The Overview view rendered with the project title, research question, three systems (C_{26}, AGL(1, 26), S_4), live Caesar example, and the educational transformation framing.
- The responsive navigation menu rendered with an accessible `Menu +` button. At the observed viewport, the menu had to be opened before selecting a page.
- Foundations rendered its group-definition explanation, modular-arithmetic example, and closure/associativity/identity/inverse labels.
- Group Explorer rendered the operation/comparison interface with C_{26}, AGL(1, 26), S_4, order comparison, and inverse/composition controls.
- The Lab rendered Caesar, affine, and S_4 system controls. Caesar encryption showed `HELLO, ALGEBRA!` becoming `KHOOR, DOJHEUD!` for key +3. The inverse control returned the plaintext visibly.
- The affine view rendered a selectable unit multiplier list (1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25), shift input, encryption/decryption controls, and the order-312 label.
- The S_4 view rendered a 24-option permutation selector, encryption/decryption controls, and the order-24 label.
- Experiments rendered its four experiment sections, including frequency/invariance, transformation inspector, composition/commutativity, and brute-force observation areas.
- The Team view rendered five profile cards, the click instruction, profile details, and the GitHub CTA linking to https://github.com/PVSSP656/final-algebra-lab.
- Team interaction: clicking a profile set `aria-expanded="true"` and `aria-controls="team-profile-project-lead"`, and revealed a matching profile-details region.
- Team interaction: clicking the open profile again set `aria-expanded="false"`.

- Team interaction: pressing Enter on another profile opened it and closed the first profile, confirming single-open switching and keyboard activation.
- Research Log rendered the research question, verified-results area, and method/limitations area.

Not yet verified in the browser

- Hover-only visual border glow was not formally measured; source inspection indicates hover styling is visual, but this remains source-supported rather than browser-measured.
- Space-key activation was not separately exercised; Enter activation was verified.
- Responsive layout at a second narrow viewport was not captured because the connected browser did not expose a viewport override in this session.
- Image fallback behaviour was not triggered.
- Affine invalid-input rejection was not exercised.
- S_4 round-trip and incomplete-final-block behaviour were not separately recorded as a browser test.
- No screenshots were saved in this run; the observations above are DOM/browser observations only.
- Console logs were not yet recorded as a separate evidence artifact.

Important navigation note

The navigation controls are responsive: at the observed viewport the menu starts closed, so direct clicks on hidden nav items do not change the view until `Menu +` is opened. This is expected responsive behaviour, not a page failure.

Evidence classification

- **Browser verified:** observations listed in the first section.
- **Present in source / pending browser evidence:** hover styling, image fallback, detailed invalid-input handling, responsive narrow layout.
- **Unresolved:** final screenshot set, second viewport, full invalid-input matrix, and console-log capture.

Next action: run the remaining targeted cases after capturing screenshots, then attach the screenshot index and any console output to this page.

Additional targeted checks — 18 October 2026

- **Space-key activation:** Browser verified. On the Team page, pressing Space on the Ananya profile opened it and exposed `aria-expanded="true"` with `aria-controls="team-profile-ananya"`.
- **S_4 forward transformation:** Browser verified for input `ABCD` and key `[2, 1, 4, 3]`; the visible encrypted result was `BADC`.
- **S_4 inverse-control observation:** The inverse control applied the inverse permutation to the current message field and displayed `BADC`, rather than automatically taking the previously displayed ciphertext as its input. This means a complete UI round-trip from ciphertext back to the original plaintext was **not established** by this interaction; record it as a behaviour requiring clarification or a separate manual workflow.
- **Affine invalid input:** The UI exposes only the 12 valid unit multipliers in its select control. No invalid multiplier can be selected through the visible control, so rejection of a non-unit multiplier remains unverified in the browser.
- **Console:** No warning or error entries were returned by the browser console check during this session.

Updated evidence boundaries

- **Browser verified:** Space-key Team activation; S_4 forward example `ABCD → BADC`; no console warnings/errors observed.
- **Unresolved or needs clarification:** S_4 UI round-trip semantics; affine invalid-input rejection; hover measurement; image fallback; narrow viewport; screenshots.

Production deployment evidence — 18 October 2026

The deployed Firebase Hosting site was tested at https://abstract-algebra-lab.web.app/.

Production page verification

All seven navigation views were reached and visually inspected:

- **00 Overview:** project title, research question, live Caesar example, and three system cards.
- **01 Foundations:** group axioms, modular arithmetic, worked Caesar example, and closure/associativity/identity/inverse explanations.
- **02 Group Explorer:** C_{26}, AGL(1, 26), S_4 cards, order comparisons, inverse and composition exploration.
- **03 The Lab:** Caesar, affine, and S_4 controls, live output, arithmetic traces, and inverse explanations.
- **04 Algebraic Structure:** shared group skeleton, three-system comparison, reversibility, composition, invariants, and limitations.
- **05 Experiments:** frequency/invariance, transformation inspector, composition/commutativity, and brute-force sections.
- **06 The Team:** five profiles, click instructions, profile details, and GitHub CTA.
- **07 Research Log:** research question, verified results, method, and limitations.

Each selected navigation button became active and the expected page content rendered.

S_4 round-trip verification

On **03 / The Lab**:

- Plaintext: `ABCD`
- Key: `[2, 1, 4, 3]`
- Encryption result: `BADC`
- Decryption result: `ABCD`
- The visible calculation trace showed block `BADC → ABCD`, inverse key `[2, 1, 4, 3]`, and `f⁻¹(f(x)) = x`.

This is now **Browser verified in production**.

Team-page verification in production

- Clicking the first profile opened it with `aria-expanded="true"`.
- `aria-controls` referenced `team-profile-project-lead`.

- Clicking the open profile collapsed it.
- Pressing Space opened a profile.
- Pressing Enter on another profile switched the single open profile and closed the first.
- GitHub CTA href verified as `https://github.com/PVSSP656/final-algebra-lab`.

Console and limitations

- No browser warnings or errors were returned during the production session.
- Hover-only border styling was not formally measured with a screenshot comparison.
- Narrow-viewport layout was not captured in this browser session.
- Image-fallback behaviour was not triggered.
- No screenshot files were captured; this remains the final evidence task for Ananya.
- The custom .com domain was not identifiable from Firebase Hosting metadata; the default Firebase URL above is the verified deployment address.

Screenshot capture update — 18 October 2026

I controlled a browser on the deployed Firebase site and captured live visual evidence for:

- Overview hero section.
- Team page at the observed narrow browser viewport.
- The Lab S_4 controls with plaintext `ABCD` and key `[2, 1, 4, 3]`.
- The deployed S_4 recovered-output panel visibly showing `ABCD`, `CIPHERTEXT (BADC)`, and S_4^{-1}.

These screenshots were displayed during the verification session but were not uploaded as persistent Notion attachments. If the paper requires downloadable image files, Ananya should save equivalent screenshots locally using the agreed naming convention.

The narrow browser viewport visibly remained usable: navigation collapsed to the menu control, the S_4 selector and action buttons remained readable, and the recovered output panel remained visible. This is browser evidence for the observed viewport, not a full breakpoint matrix.

Hover-only border glow and image fallback were not separately captured.

Attached screenshot evidence

A FIELD GUIDE TO REVERSIBLE SYSTEMS ◈

Cryptography *through* abstract algebra.



An interactive study of three transformation systems, where every click is a calculation and every inverse has a reason.

Enter the laboratory ↗

01 / RESEARCH QUESTION

How does abstract algebra help us understand, compare, and verify reversible transformation systems used in classical cryptography?

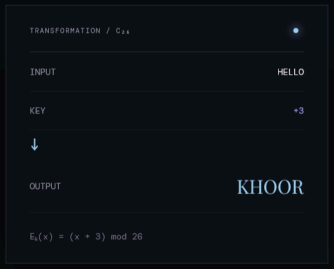
READ THE PREMISE ↗

LIVE SIGNAL

Every output has a trail.

Watch a Caesar shift operate modulo 26. The engine is live: change the key, then trace the inverse.

OPEN INTERACTIVE LAB →



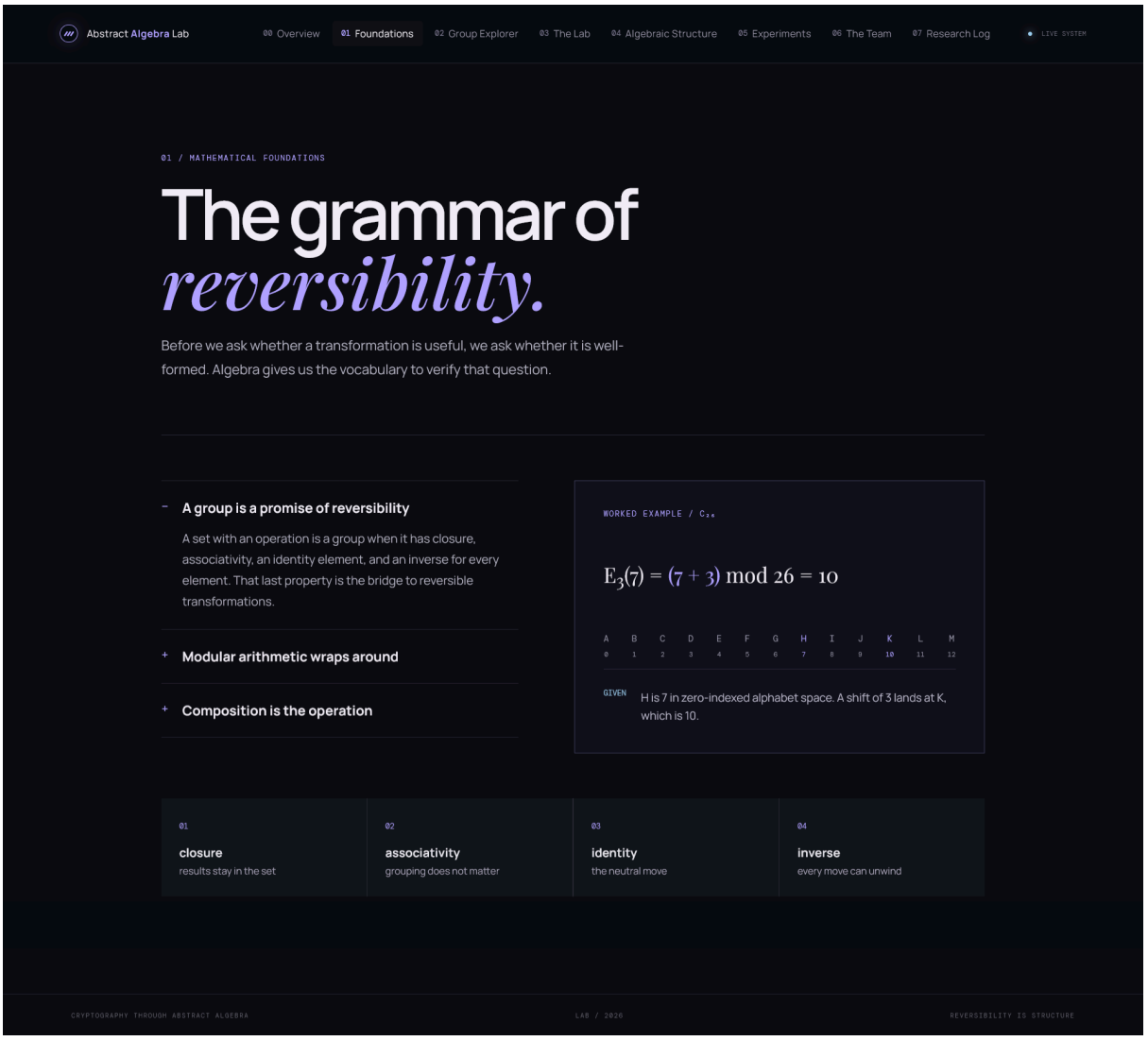
THREE SYSTEMS / ONE QUESTION

Compare the structure.

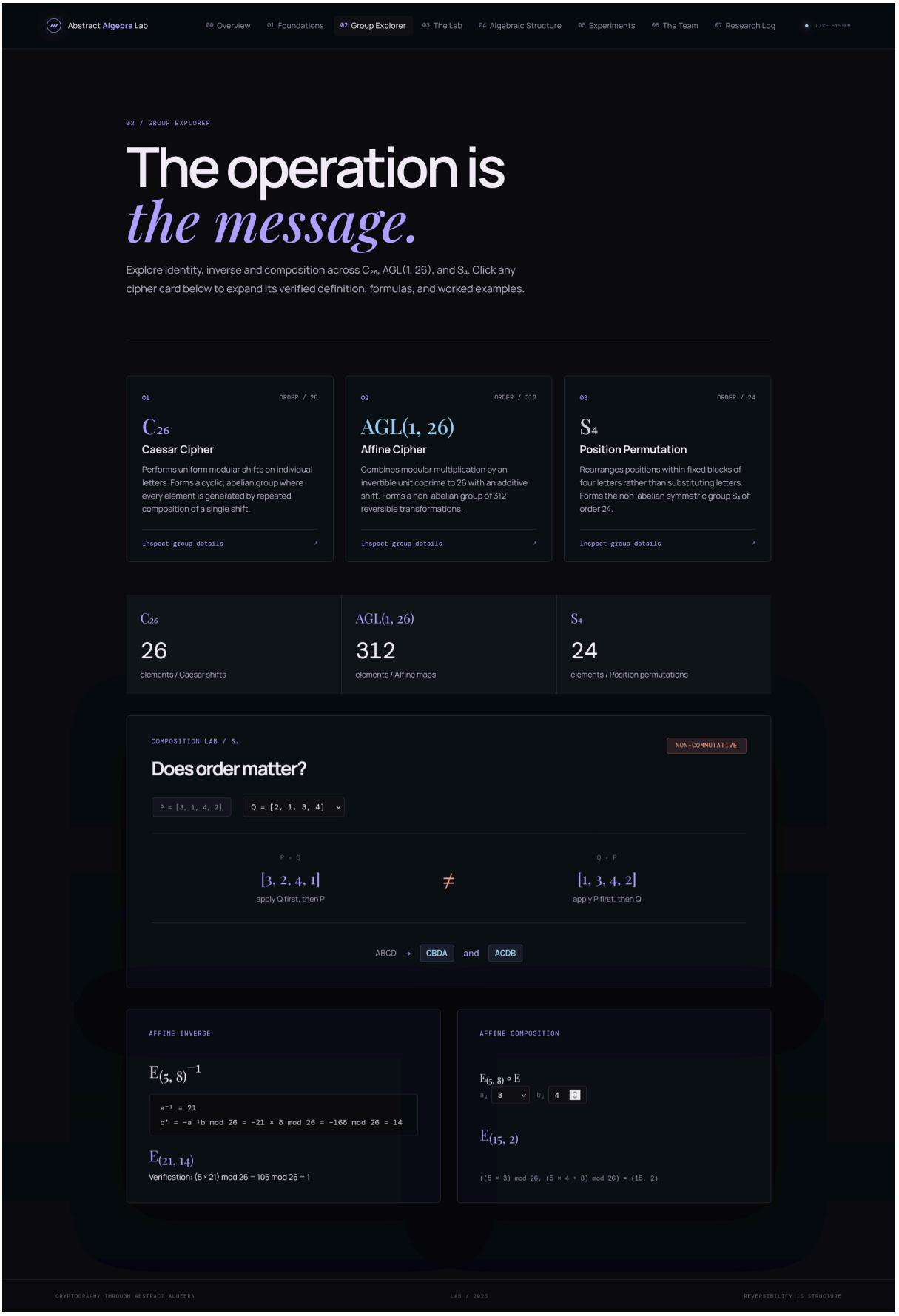
EXPLORE GROUPS →

01	02	03
C_{26}	AGL(1, 26)	S_4
Caesar shifts	**Affine maps**	**Position permutations**
order / 26	order / 312	order / 24
↗	↗	↗

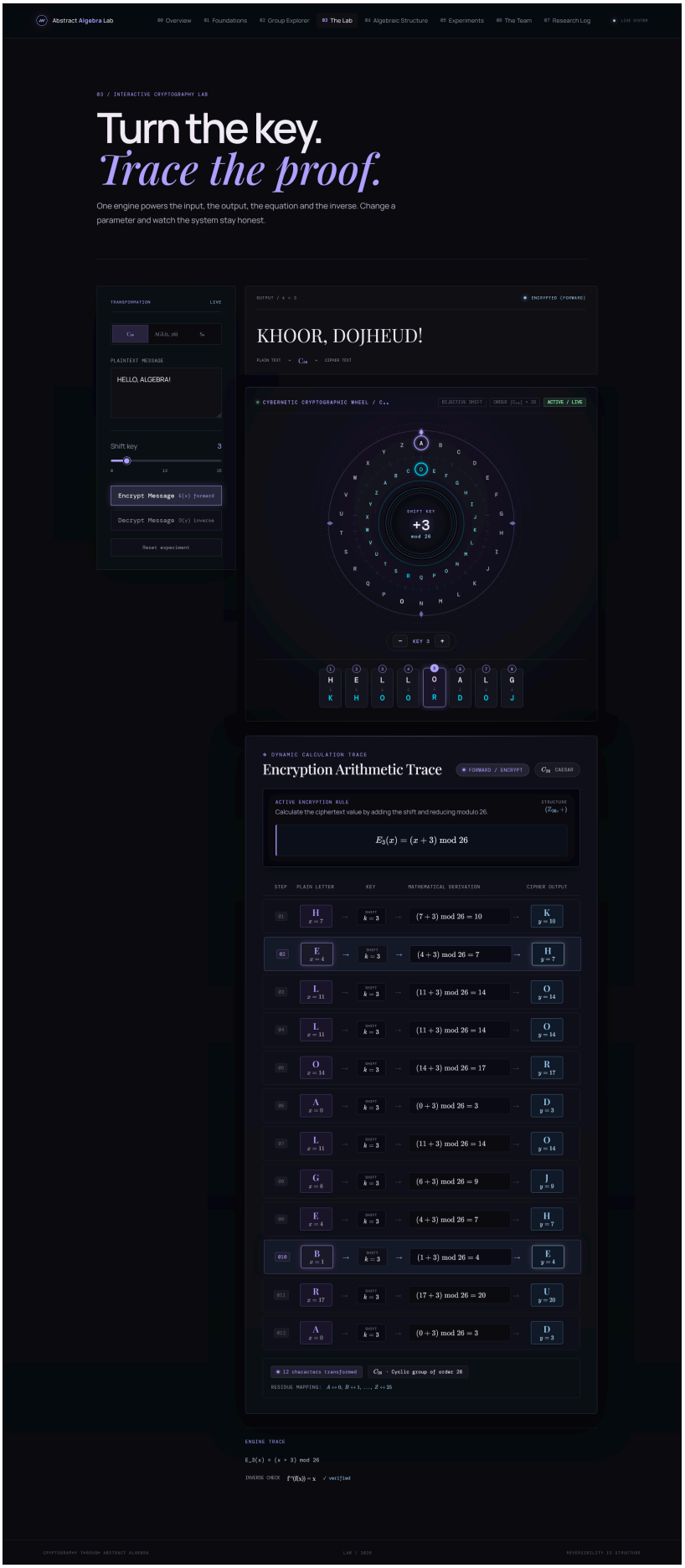
Attached file: `ANANYA_01_overview_2026-10-18.png`



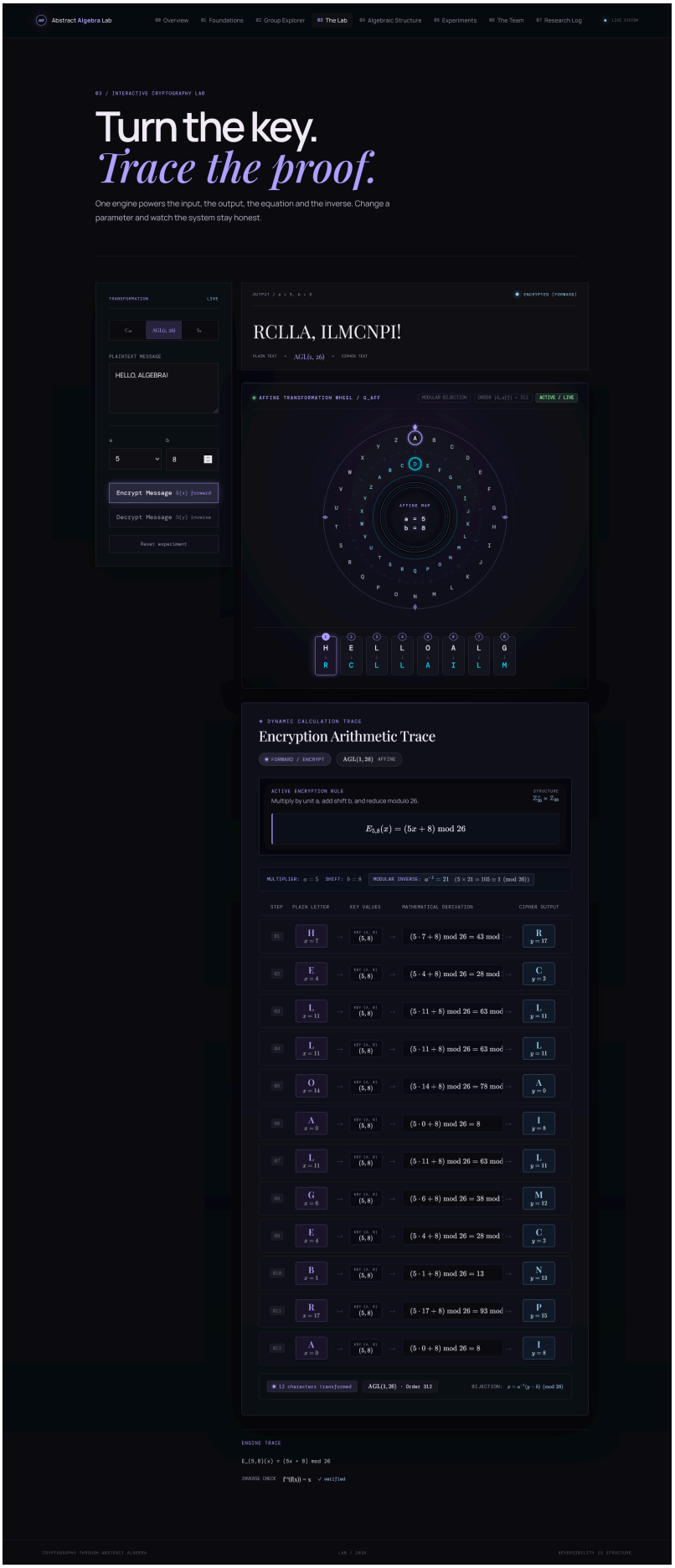
Attached file: `ANANYA_02_foundations_2026-10-18.png`



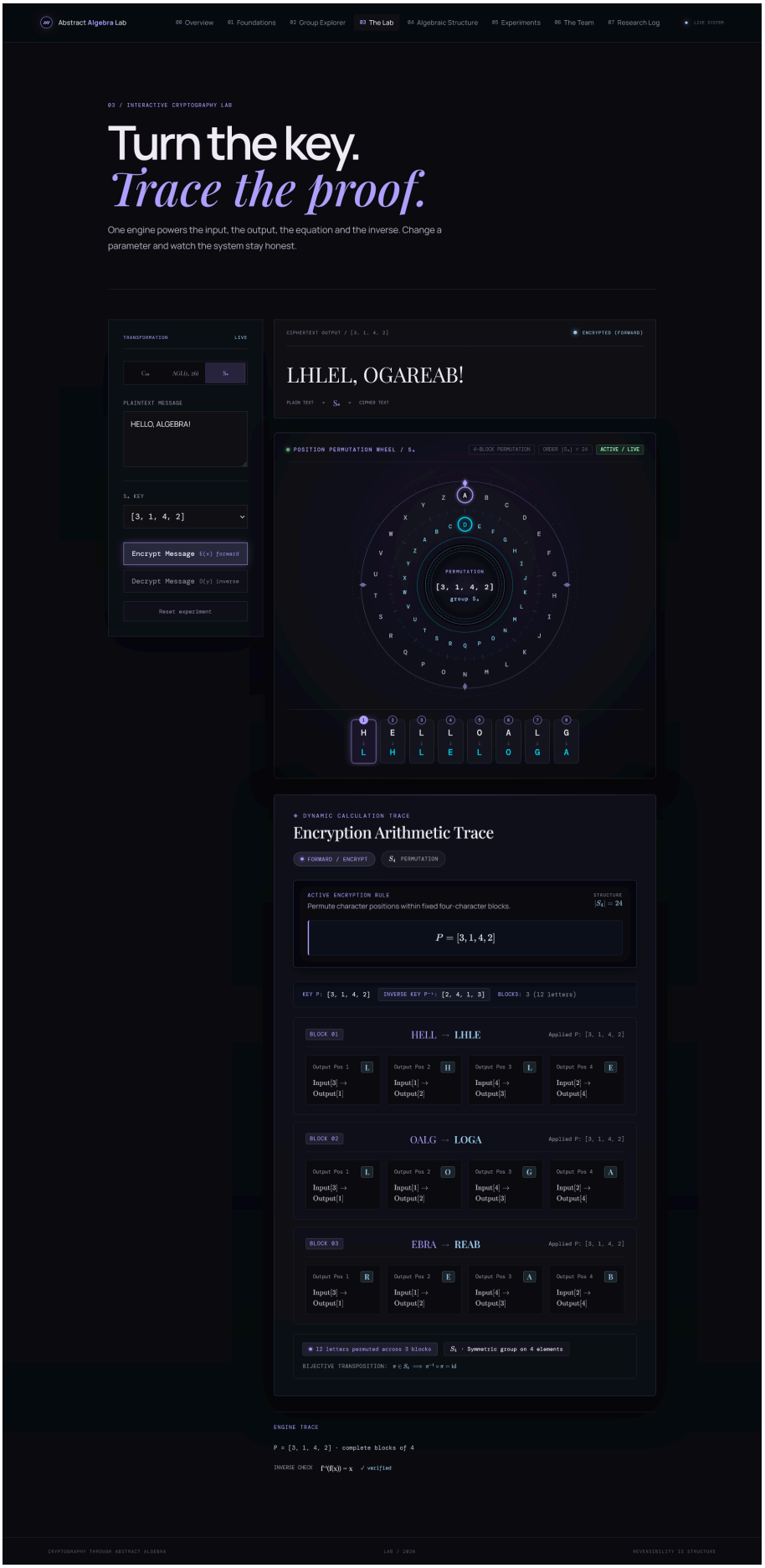
Attached file: `ANANYA_03_group-explorer_2026-10-18.png`


Abstract Algebra Lab
00 Overview
01 Foundations
02 Group Explorer
03 The Lab
04 Algebraic Structure
05 Experiments
06 The Team
07 Research Log
03 / INTERACTIVE CRYPTOGRAPHY LAB
Turn the key.
Trace the proof.
One engine powers the input, the output, the equation and the inverse. Change a parameter and watch the system stay honest.
TRANSFORMATION
LIVE
PLAINTEXT MESSAGE
HELLO, ALGEBRA!
Shift key
3
Encrypt Message E(x) forward
Decrypt Message D(y) inverse
Reset experiment
OUTPUT / k = 3
ENCRYPTED (FORWARD)
KHOOR, DOJHEUD!
PLAIN TEXT
CIPHER TEXT
CYBERNETIC CRYPTOGRAPHIC WHEEL / C26
ACTIVE / LIVE
SHIFT KEY
+3
mod 26
KEY 3
H E L L O A L G
K H O O R D O J
DYNAMIC CALCULATION TRACE
Encryption Arithmetic Trace
FORWARD / ENCRYPT
C26 CAESAR
ACTIVE ENCRYPTION RULE
Calculate the ciphertext value by adding the shift and reducing modulo 26.
STRUCTURE
$E_3(x) = (x + 3) \bmod 26$
STEP
PLAIN LETTER
KEY
MATHEMATICAL DERIVATION
CIPHER OUTPUT
01 H x = 7 → k = 3 → (7 + 3) mod 26 = 10 → K y = 10
02 E x = 4 → k = 3 → (4 + 3) mod 26 = 7 → H y = 7
03 L x = 11 → k = 3 → (11 + 3) mod 26 = 14 → O y = 14
04 L x = 11 → k = 3 → (11 + 3) mod 26 = 14 → O y = 14
05 O x = 14 → k = 3 → (14 + 3) mod 26 = 17 → R y = 17
06 A x = 0 → k = 3 → (0 + 3) mod 26 = 3 → D y = 3
07 L x = 11 → k = 3 → (11 + 3) mod 26 = 14 → O y = 14
08 G x = 6 → k = 3 → (6 + 3) mod 26 = 9 → J y = 9
09 E x = 4 → k = 3 → (4 + 3) mod 26 = 7 → H y = 7
010 B x = 1 → k = 3 → (1 + 3) mod 26 = 4 → E y = 4
011 R x = 17 → k = 3 → (17 + 3) mod 26 = 20 → U y = 20
012 A x = 0 → k = 3 → (0 + 3) mod 26 = 3 → D y = 3
12 characters transformed
C26 · Cyclic group of order 26
RESIDUE MAPPING: A ↦ 0, B ↦ 1, …, Z ↦ 25
ENGINE TRACE
E_3(x) = (x + 3) mod 26
INVERSE CHECK
✓ verified

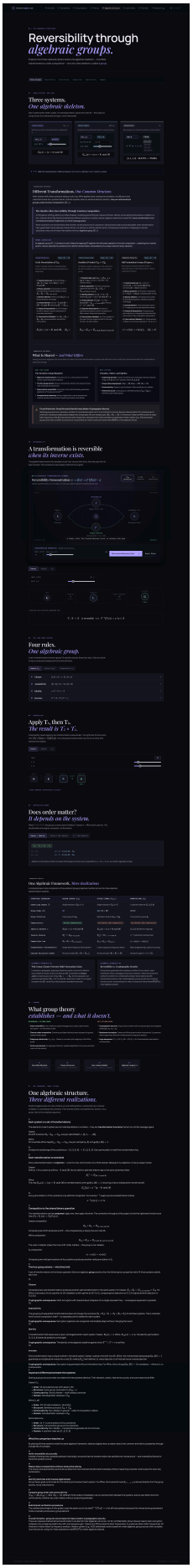
Attached file: `ANANYA_04_lab-caesar_2026-10-18.png`


Abstract Algebra Lab
00 Overview
01 Foundations
02 Group Explorer
03 The Lab
04 Algebraic Structure
05 Experiments
06 The Team
07 Research Log
03 / INTERACTIVE CRYPTOGRAPHY LAB
Turn the key.
Trace the proof.
One engine powers the input, the output, the equation and the inverse. Change a parameter and watch the system stay honest.
TRANSFORMATION
LIVE
PLAINTEXT MESSAGE
HELLO, ALGEBRA!
a
b
5
8
Encrypt Message E(x) forward
Decrypt Message D(y) inverse
Reset experiment
ENCRYPTED (FORWARD)
RCLLA, ILMCNPI!
PLAIN TEXT
CIPHER TEXT
AFFINE TRANSFORMATION WHEEL / G_AFF
MODULAR BIJECTION
ACTIVE / LIVE
AFFINE MAP
a = 5
b = 8
H E L L O A L G
R C L L A I L M
DYNAMIC CALCULATION TRACE
Encryption Arithmetic Trace
FORWARD / ENCRYPT
AGL(1, 26) AFFINE
ACTIVE ENCRYPTION RULE
Multiply by unit a, add shift b, and reduce modulo 26.
STRUCTURE
$E_{5,8}(x) = (5x + 8) \bmod 26$
MULTIPLIER: a = 5
SHIFT: b = 8
MODULAR INVERSE: $a^{-1} = 21$ $(5 \times 21 = 105 \equiv 1 \pmod{26})$
STEP
PLAIN LETTER
KEY VALUES
MATHEMATICAL DERIVATION
CIPHER OUTPUT
01 H x = 7 → KEY (a, b) (5, 8) → (5 · 7 + 8) mod 26 = 43 mod → R y = 17
02 E x = 4 → KEY (a, b) (5, 8) → (5 · 4 + 8) mod 26 = 28 mod → C y = 2
03 L x = 11 → KEY (a, b) (5, 8) → (5 · 11 + 8) mod 26 = 63 mod → L y = 11
04 L x = 11 → KEY (a, b) (5, 8) → (5 · 11 + 8) mod 26 = 63 mod → L y = 11
05 O x = 14 → KEY (a, b) (5, 8) → (5 · 14 + 8) mod 26 = 78 mod → A y = 0
06 A x = 0 → KEY (a, b) (5, 8) → (5 · 0 + 8) mod 26 = 8 → I y = 8
07 L x = 11 → KEY (a, b) (5, 8) → (5 · 11 + 8) mod 26 = 63 mod → L y = 11
08 G x = 6 → KEY (a, b) (5, 8) → (5 · 6 + 8) mod 26 = 38 mod → M y = 12
09 E x = 4 → KEY (a, b) (5, 8) → (5 · 4 + 8) mod 26 = 28 mod → C y = 2
010 B x = 1 → KEY (a, b) (5, 8) → (5 · 1 + 8) mod 26 = 13 → N y = 13
011 R x = 17 → KEY (a, b) (5, 8) → (5 · 17 + 8) mod 26 = 93 mod → P y = 15
012 A x = 0 → KEY (a, b) (5, 8) → (5 · 0 + 8) mod 26 = 8 → I y = 8
12 characters transformed
AGL(1, 26) · Order 312
BIJECTION: $x = a^{-1}(y - b) \pmod{26}$
ENGINE TRACE
E_(5,8)(x) = (5x + 8) mod 26
INVERSE CHECK f^(-1)(f(x)) = x
✓ verified

Attached file: `ANANYA_05_lab-affine_2026-10-18.png`


Abstract Algebra Lab
00 Overview
01 Foundations
02 Group Explorer
03 The Lab
04 Algebraic Structure
05 Experiments
06 The Team
07 Research Log
LIVE SYSTEM
03 / INTERACTIVE CRYPTOGRAPHY LAB
Turn the key.
Trace the proof.
One engine powers the input, the output, the equation and the inverse. Change a parameter and watch the system stay honest.
TRANSFORMATION
LIVE
PLAINTEXT MESSAGE
HELLO, ALGEBRA!
S₄ KEY
[3, 1, 4, 2]
Encrypt Message E(x) forward
Decrypt Message D(y) inverse
Reset experiment
CIPHERTEXT OUTPUT / [3, 1, 4, 2]
ENCRYPTED (FORWARD)
LHLEL, OGAREAB!
PLAIN TEXT
CIPHER TEXT
POSITION PERMUTATION WHEEL / S₄
4-BLOCK PERMUTATION
ORDER |S₄| = 24
ACTIVE / LIVE
PERMUTATION
[3, 1, 4, 2]
group S₄
DYNAMIC CALCULATION TRACE
Encryption Arithmetic Trace
FORWARD / ENCRYPT
S₄ PERMUTATION
ACTIVE ENCRYPTION RULE
Permute character positions within fixed four-character blocks.
STRUCTURE
|S₄| = 24
P = [3, 1, 4, 2]
KEY P: [3, 1, 4, 2]
INVERSE KEY P⁻¹: [2, 4, 1, 3]
BLOCKS: 3 (12 letters)
BLOCK 01
HELL → LHLE
Applied P: [3, 1, 4, 2]
BLOCK 02
OALG → LOGA
Applied P: [3, 1, 4, 2]
BLOCK 03
EBRA → REAB
Applied P: [3, 1, 4, 2]
12 letters permuted across 3 blocks
S₄ · Symmetric group on 4 elements
BIJECTIVE TRANSPOSITION: π ∈ S₄ ⟹ π⁻¹ ∘ π = id
ENGINE TRACE
P = [3, 1, 4, 2] · complete blocks of 4
INVERSE CHECK f⁻¹(f(x)) = x ✓ verified
CRYPTOGRAPHY THROUGH ABSTRACT ALGEBRA
LAB / 2026
REVERSIBILITY IS STRUCTURE

Attached file: `ANANYA_06_lab-s4-encryption_2026-10-18.png`



Reversibility through *algebraic groups.*

Three systems. *One algebraic skeleton.*

Different Transformations, *One Common Structure.*

What Is Shared — *And What Differs*

A transformation is reversible *when its inverse exists.*

Four rules. *One algebraic group.*

Apply T_1, then T_2. *The result is $T_3 \circ T_2$.*

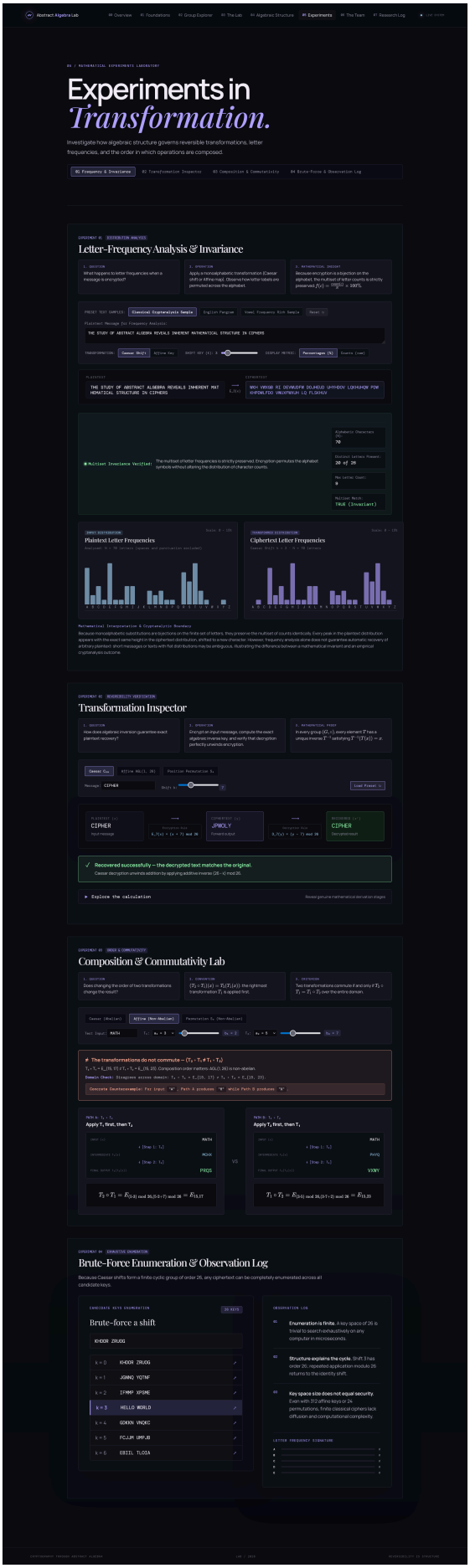
Does order matter? *It depends on the system.*

One Algebraic Framework. *Many Realizations*

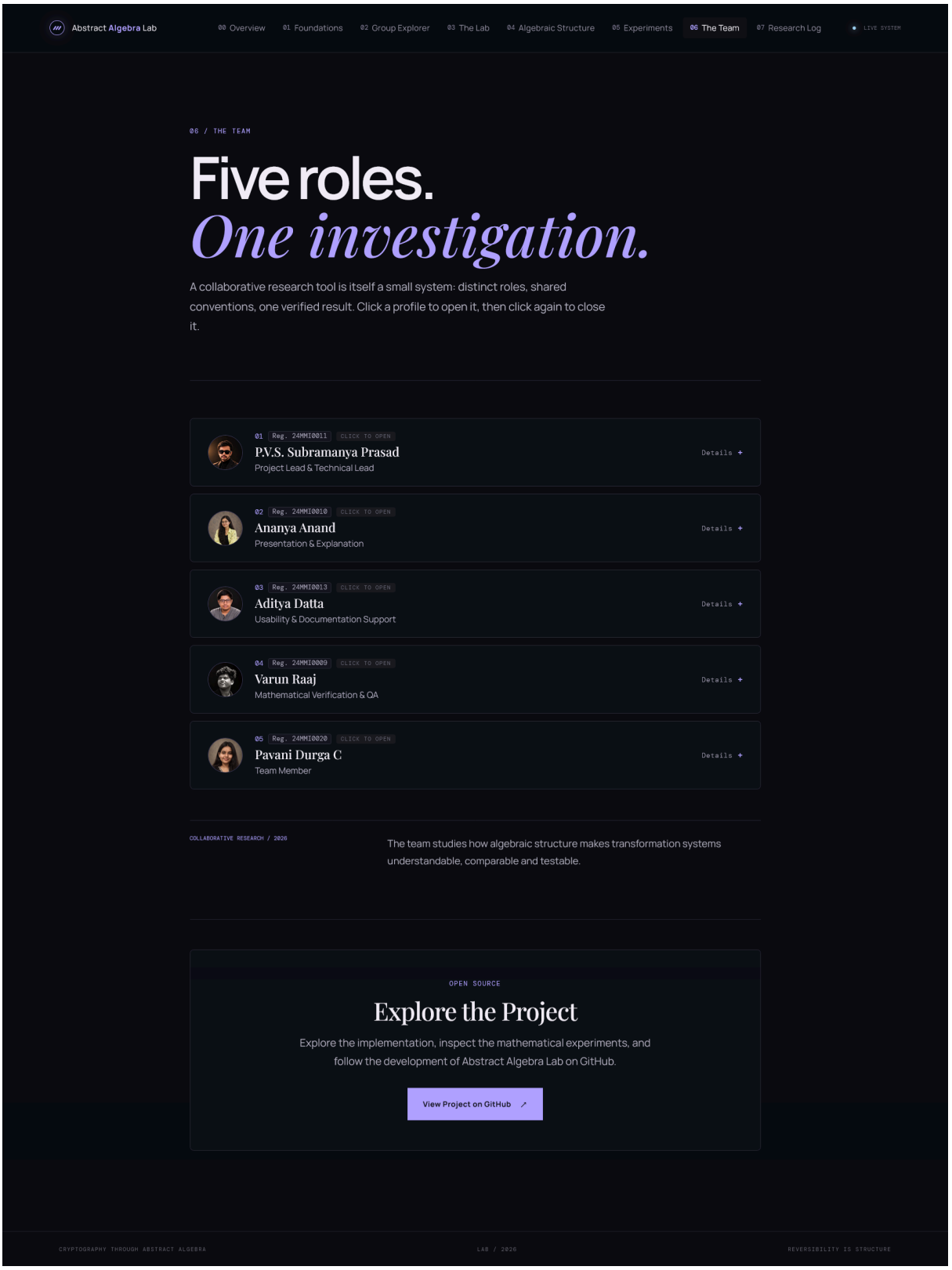
What group theory *establishes — and what it doesn't.*

One algebraic structure. *Three different realizations.*

Attached file: `ANANYA_08_algebraic-structure_2026-10-18.png`


Abstract Algebra Lab
Overview
Foundations
Group Explorer
The Lab
Algebraic Structure
Experiments
The Team
Research Log
Live Demo
05 / MATHEMATICAL EXPERIMENTS LABORATORY
Experiments in
Transformation.
Investigate how algebraic structure governs reversible transformations, letter frequencies, and the order in which operations are composed.
01 Frequency & Invariance
02 Transformation Inspector
03 Composition & Commutativity
04 Brute-Force & Observation Log
EXPERIMENT 01 DISTRIBUTION ANALYSIS
Letter-Frequency Analysis & Invariance
1 · QUESTION
What happens to letter frequencies when a message is encrypted?
2 · OPERATION
Apply a monoalphabetic transformation (Caesar shift or Affine map). Observe how letter labels are permuted across the alphabet.
3 · MATHEMATICAL INSIGHT
Because encryption is a bijection on the alphabet, the multiset of letter counts is strictly preserved.
PRESET TEST SAMPLES:
Classical Cryptanalysis Sample
English Pangram
Vowel-Frequency Rich Sample
Reset ↻
Plaintext Message (to Frequency Analyze):
THE STUDY OF ABSTRACT ALGEBRA REVEALS INHERENT MATHEMATICAL STRUCTURE IN CIPHERS
TRANSFORMATION:
Caesar Shift
Affine Key
SHIFT KEY (K): 3
DISPLAY METRIC:
Percentages (%)
Counts (raw)
PLAINTEXT
THE STUDY OF ABSTRACT ALGEBRA REVEALS INHERENT MAT HEMATICAL STRUCTURE IN CIPHERS
E_3(x)
CIPHERTEXT
WKH VWXGB RI DEVWUDFW DOJHEUD UHYHDOV LQKHUHQW PDW KHPDWLFDO VWUXFWXUH LQ FLSKHUV
Multiset Invariance Verified:
The multiset of letter frequencies is strictly preserved. Encryption permutes the alphabet symbols without altering the distribution of character counts.
Alphabetic Characters (N):
70
Distinct Letters Present:
20 of 26
Max Letter Count:
9
Multiset Match:
TRUE (Invariant)
INPUT DISTRIBUTION
Plaintext Letter Frequencies
Analysed: N = 70 letters (spaces and punctuation excluded)
TRANSFORMED DISTRIBUTION
Ciphertext Letter Frequencies
Caesar Shift k = 3 · N = 70 letters
Mathematical Interpretation & Cryptanalytic Boundary
Because monoalphabetic substitutions are bijections on the finite set of letters, they preserve the multiset of counts identically. Every peak in the plaintext distribution appears with the exact same height in the ciphertext distribution, shifted to a new character. However, frequency analysis alone does not guarantee automatic recovery of arbitrary plaintext: short messages or texts with flat distributions may be ambiguous, illustrating the difference between a mathematical invariant and an empirical cryptanalysis outcome.
EXPERIMENT 02 REVERSIBILITY VERIFICATION
Transformation Inspector
1 · QUESTION
How does algebraic inversion guarantee exact plaintext recovery?
2 · OPERATION
Encrypt an input message, compute the exact algebraic inverse key, and verify that decryption perfectly unwinds encryption.
3 · MATHEMATICAL PROOF
In every group $(G, \cdot)$, every element T has a unique inverse T^{-1} satisfying $T^{-1}(T(x)) = x$.
Caesar Cu
Affine Mix(5, 20)
Position Permutation π_4
Message: CIPHER
Shift k
Load Preset ↻
PLAINTEXT (x)
CIPHER
Input message
ENCRYPTION E_k(x) = (x + 7) mod 26
CIPHERTEXT (y)
JPNOLY
Forward output
DECRYPTION D_k(y) = (y − 7) mod 26
DECRYPTED (x')
CIPHER
Decrypted result
✓ Recovered successfully — the decrypted text matches the original.
Caesar decryption unwinds addition by applying additive inverse 026 − 0 mod 26.
▶ Explore the calculation
Reveal genuine mathematical derivation images
EXPERIMENT 03 ORDER & COMMUTATIVITY
Composition & Commutativity Lab
1 · QUESTION
Does changing the order of two transformations change the result?
2 · CONVENTION
$(T_2 \circ T_1)(x) = T_2(T_1(x))$: the rightmost transformation T_1 is applied first.
3 · CRITERION
Two transformations commute if and only if $T_2 \circ T_1 = T_1 \circ T_2$ over the entire domain.
Caesar (Abelian)
Affine (Non-Abelian)
Permutation π_4 (Non-Abelian)
Test Input: MATH
T_1: $a_1 = 3$
$b_1 = 2$
T_2: $a_2 = 5$
$b_2 = 7$
≠ The transformations do not commute — $(T_2 \circ T_1 \neq T_1 \circ T_2)$
$T_2 \circ T_1 = E_{15}$, $T_1 \circ T_2 = E_{19}$. 20. Composition order matters: $AB \neq BA$ in non-abelian.
Domain Check: Disagrees across domain: $T_2 \circ T_1 = E_{15}$, 17 ≠ $T_1 \circ T_2 = E_{19}$, 20).
Concrete Counterexample: For input "A", Path A produces "F" while Path B produces "N".
PATH A: $T_1 \to T_2$
Apply T_1 first, then T_2
MATH
+ [Step 1: T_1]
MDJK
+ [Step 2: T_2]
PRQS
$T_2 \circ T_1 = E_{(5,0)} \bmod 26, (5 \cdot 3 + 7) \bmod 26 = E_{15,17}$
VS
PATH B: $T_2 \to T_1$
Apply T_2 first, then T_1
MATH
+ [Step 1: T_2]
PAYQ
+ [Step 2: T_1]
VXNY
$T_1 \circ T_2 = E_{(3,6)} \bmod 26, (5 \cdot 7 + 0) \bmod 26 = E_{11,23}$
EXPERIMENT 04 EXHAUSTIVE EXPLORATION
Brute-Force Enumeration & Observation Log
Because Caesar shifts form a finite cyclic group of order 26, any ciphertext can be completely enumerated across all candidate keys.
CANDIDATE KEYS ENUMERATION
26 KEYS
Brute-force a shift
KHOOR ZRUGG
k = 0 KHOOR ZRUGG
k = 1 JGNNQ YQTNF
k = 2 IFMMP XPSME
k = 3 HELLO WORLD
k = 4 GDKKN VNQKC
k = 5 FCJJM UMPJB
k = 6 EBIIL TLOIA
OBSERVATION LOG
01 Enumeration is finite. A key space of 26 is trivial to search exhaustively on any computer in microseconds.
02 Structure explains the cycle. Shift 3 has order 26: repeated application modulo 26 returns to the identity shift.
03 Key space size does not equal security. Even with 312 effective keys or 24 permutations, finite classical ciphers lack diffusion and computational complexity.
LETTER FREQUENCY SIGNATURE

Attached file: `ANANYA_09_experiments_2026-10-18.png`



Attached file: `ANANYA_10_team-collapsed_2026-10-18.png`

06 / THE TEAM

Five roles. *One investigation.*

A collaborative research tool is itself a small system: distinct roles, shared conventions, one verified result. Click a profile to open it, then click again to close it.

01 Reg. 24MMI0011 CLICK TO OPEN
P.V.S. Subramanya Prasad
Project Lead & Technical Lead
Details +

02 Reg. 24MMI0010 OPEN
Ananya Anand
Presentation & Explanation
Close –



RESEARCH PROFILE / 02
Ananya Anand
Registration No. · 24MMI0010
Presentation & Explanation

KEY RESPONSIBILITIES & SCOPE
→ Presentation and plain-language explanations.
→ Demonstration script and viva preparation.

03 Reg. 24MMI0013 CLICK TO OPEN
Aditya Datta
Usability & Documentation Support
Details +

04 Reg. 24MMI0009 CLICK TO OPEN
Varun Raaj
Mathematical Verification & QA
Details +

05 Reg. 24MMI0020 CLICK TO OPEN
Pavani Durga C
Team Member
Details +

COLLABORATIVE RESEARCH / 2026

The team studies how algebraic structure makes transformation systems understandable, comparable and testable.

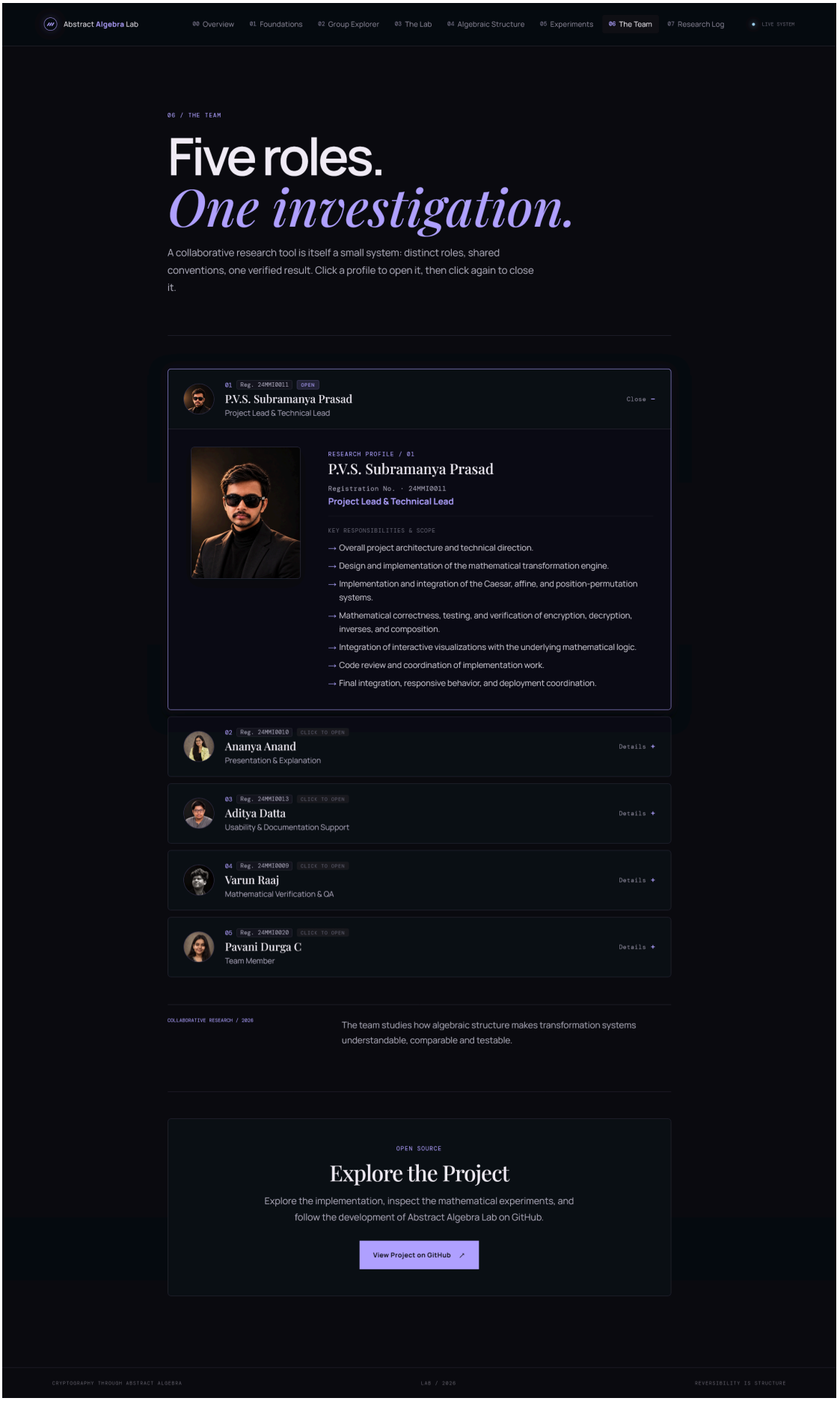
OPEN SOURCE

Explore the Project

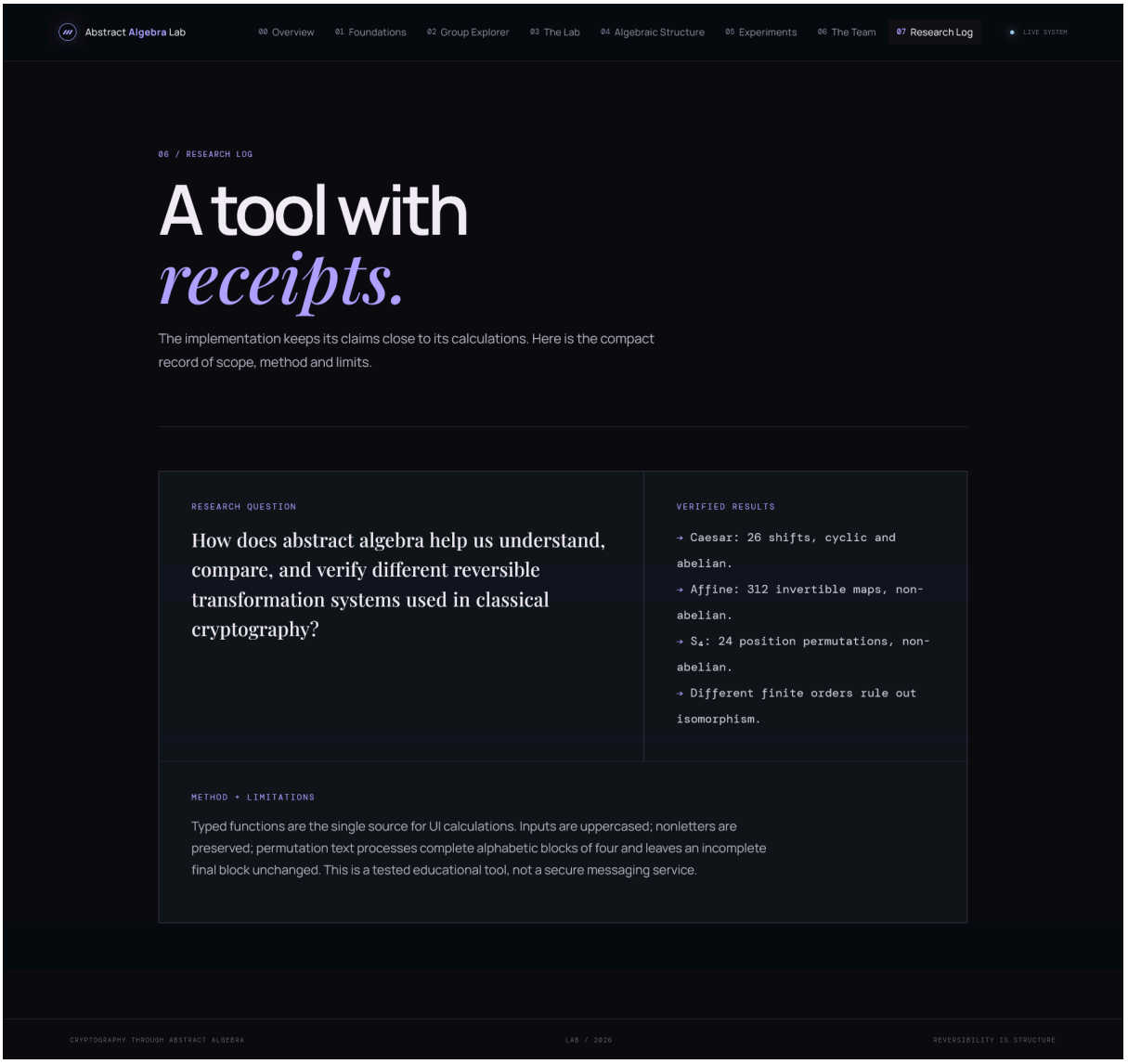
Explore the implementation, inspect the mathematical experiments, and follow the development of Abstract Algebra Lab on GitHub.

View Project on GitHub ↗

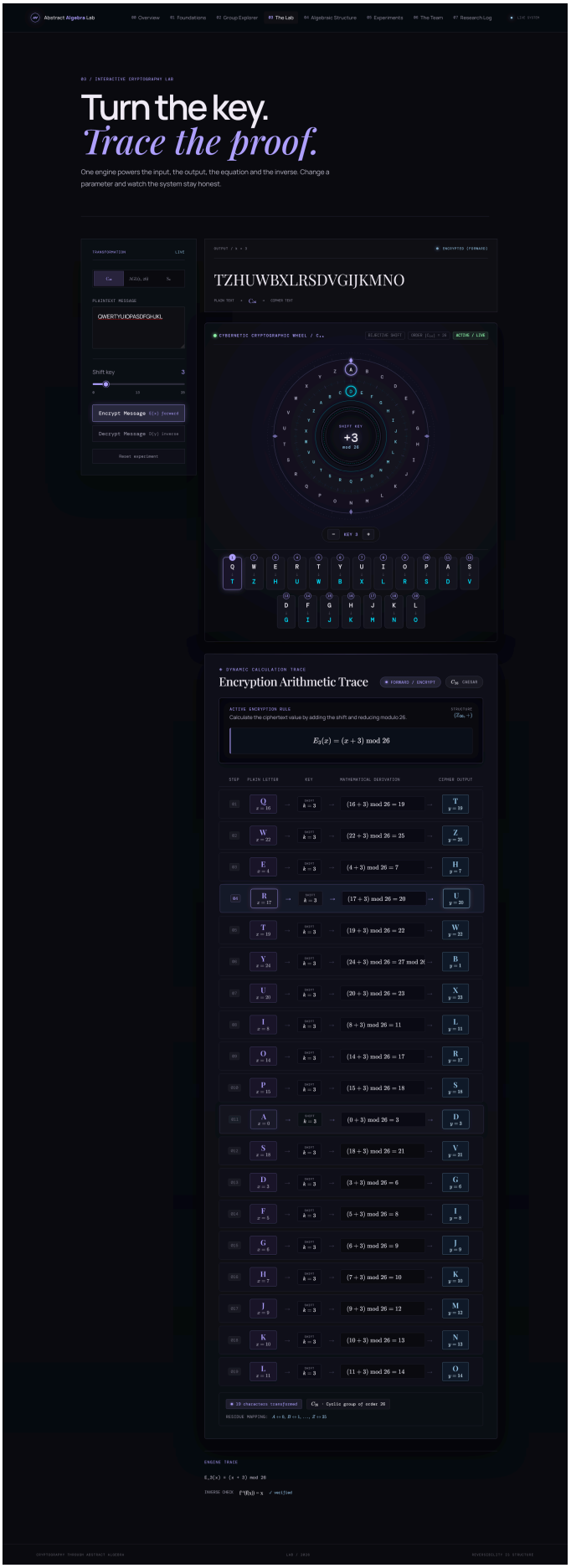
Attached file: `ANANYA_11_team-expanded_2026-10-18.png`


Abstract Algebra Lab
00 Overview
01 Foundations
02 Group Explorer
03 The Lab
04 Algebraic Structure
05 Experiments
06 The Team
07 Research Log
LIVE SYSTEM
06 / THE TEAM
Five roles.
One investigation.
A collaborative research tool is itself a small system: distinct roles, shared conventions, one verified result. Click a profile to open it, then click again to close it.
01 Reg. 24MMI0011 OPEN
P.V.S. Subramanya Prasad
Project Lead & Technical Lead
Close –
RESEARCH PROFILE / 01
P.V.S. Subramanya Prasad
Registration No. · 24MMI0011
Project Lead & Technical Lead
KEY RESPONSIBILITIES & SCOPE
→ Overall project architecture and technical direction.
→ Design and implementation of the mathematical transformation engine.
→ Implementation and integration of the Caesar, affine, and position-permutation systems.
→ Mathematical correctness, testing, and verification of encryption, decryption, inverses, and composition.
→ Integration of interactive visualizations with the underlying mathematical logic.
→ Code review and coordination of implementation work.
→ Final integration, responsive behavior, and deployment coordination.
02 Reg. 24MMI0010 CLICK TO OPEN
Ananya Anand
Presentation & Explanation
Details +
03 Reg. 24MMI0013 CLICK TO OPEN
Aditya Datta
Usability & Documentation Support
Details +
04 Reg. 24MMI0009 CLICK TO OPEN
Varun Raaj
Mathematical Verification & QA
Details +
05 Reg. 24MMI0020 CLICK TO OPEN
Pavani Durga C
Team Member
Details +
COLLABORATIVE RESEARCH / 2026
The team studies how algebraic structure makes transformation systems understandable, comparable and testable.
OPEN SOURCE
Explore the Project
Explore the implementation, inspect the mathematical experiments, and follow the development of Abstract Algebra Lab on GitHub.
View Project on GitHub ↗
CRYPTOGRAPHY THROUGH ABSTRACT ALGEBRA
LAB / 2026
REVERSIBILITY IS STRUCTURE

Attached file: `ANANYA_12_team-switching_2026-10-18.png`



Attached file: `ANANYA_13_research-log_2026-10-18.png`


Abstract Algebra Lab
00 Overview
01 Foundations
02 Group Explorer
03 The Lab
04 Algebraic Structure
05 Experiments
06 The Team
07 Research Log
03 / INTERACTIVE CRYPTOGRAPHY LAB
Turn the key.
Trace the proof.
One engine powers the input, the output, the equation and the inverse. Change a parameter and watch the system stay honest.
TRANSFORMATION
LIVE
PLAINTEXT MESSAGE
QWERTYUIOPASDFGHJKL
Shift key
3
Encrypt Message
Decrypt Message
Reset experiment
ENCRYPTED (FORWARD)
TZHUWBXLRSDVGIJKMNO
CYBERNETIC CRYPTOGRAPHIC WHEEL / C26
ACTIVE / LIVE
SHIFT KEY
+3
mod 26
KEY 3
DYNAMIC CALCULATION TRACE
Encryption Arithmetic Trace
FORWARD / ENCRYPT
ACTIVE ENCRYPTION RULE
Calculate the ciphertext value by adding the shift and reducing modulo 26.
$$E_3(x) = (x + 3) \bmod 26$$
STEP
PLAIN LETTER
KEY
MATHEMATICAL DERIVATION
CIPHER OUTPUT
01 Q x = 16 shift k = 3 (16 + 3) mod 26 = 19 T y = 19
02 W x = 22 shift k = 3 (22 + 3) mod 26 = 25 Z y = 25
03 E x = 4 shift k = 3 (4 + 3) mod 26 = 7 H y = 7
04 R x = 17 shift k = 3 (17 + 3) mod 26 = 20 U y = 20
05 T x = 19 shift k = 3 (19 + 3) mod 26 = 22 W y = 22
06 Y x = 24 shift k = 3 (24 + 3) mod 26 = 27 mod 26 B y = 1
07 U x = 20 shift k = 3 (20 + 3) mod 26 = 23 X y = 23
08 I x = 8 shift k = 3 (8 + 3) mod 26 = 11 L y = 11
09 O x = 14 shift k = 3 (14 + 3) mod 26 = 17 R y = 17
010 P x = 15 shift k = 3 (15 + 3) mod 26 = 18 S y = 18
011 A x = 0 shift k = 3 (0 + 3) mod 26 = 3 D y = 3
012 S x = 18 shift k = 3 (18 + 3) mod 26 = 21 V y = 21
013 D x = 3 shift k = 3 (3 + 3) mod 26 = 6 G y = 6
014 F x = 5 shift k = 3 (5 + 3) mod 26 = 8 I y = 8
015 G x = 6 shift k = 3 (6 + 3) mod 26 = 9 J y = 9
016 H x = 7 shift k = 3 (7 + 3) mod 26 = 10 K y = 10
017 J x = 9 shift k = 3 (9 + 3) mod 26 = 12 M y = 12
018 K x = 10 shift k = 3 (10 + 3) mod 26 = 13 N y = 13
019 L x = 11 shift k = 3 (11 + 3) mod 26 = 14 O y = 14
19 characters transformed
Cyclic group of order 26
ENGINE TRACE
$E_3(x) = (x + 3) \bmod 26$
INVERSE CHECK
$D^{-1}(E(x)) = x$ ✓ verified

Attached file: `ANANYA_14_dynamic-unique-mapping-9plus_2026-10-18.png`

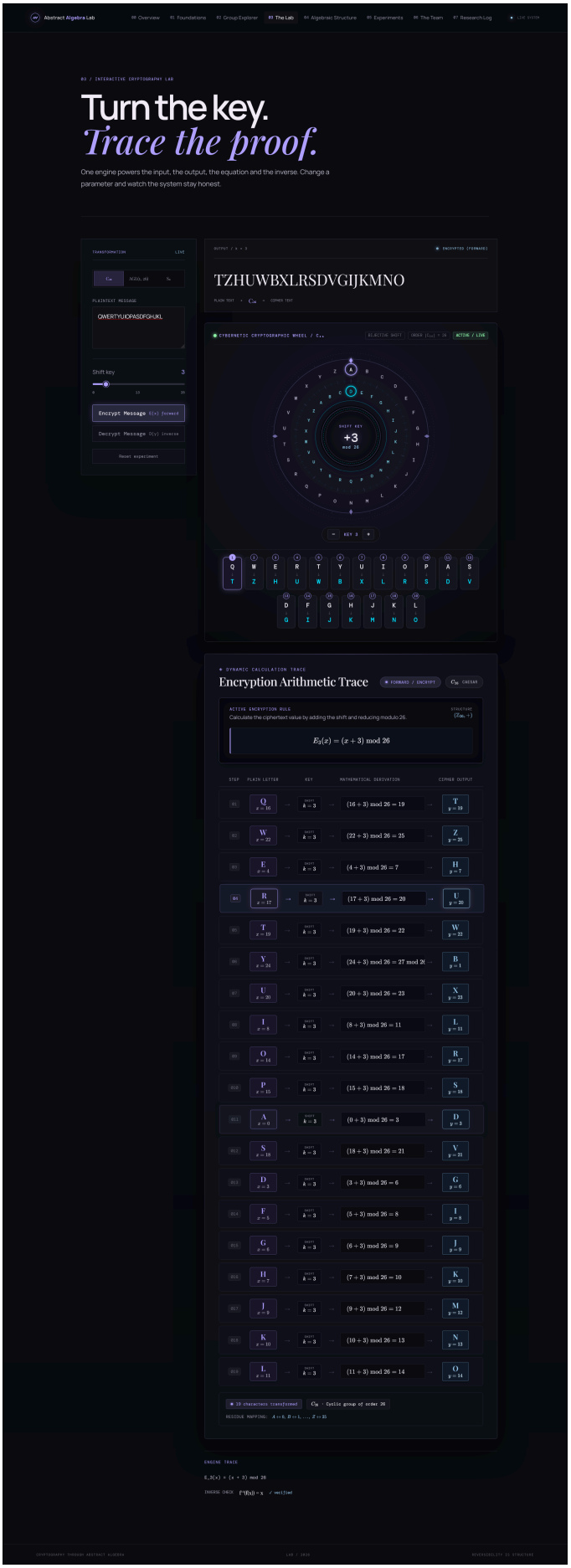
Dynamic unique-letter mapping update — 18 October 2026

The eight-card limitation has been fixed and deployed. The letter-mapping visualisation now renders every distinct alphabetic character in first-appearance order, including 9 or more unique letters, and the layout wraps instead of truncating the mapping cards.

Verification evidence:

- `npm run lint` passed.
- `npm run verify` passed.
- `npm run build` passed.
- GitHub commit: `91fb67b13a47bf62ca2e7dedc348ad69a6863e66`.
- Firebase deployment verified at https://abstract-algebra-lab.web.app.

A dedicated 9+ unique-letter screenshot is not currently present in `evidence/screenshots`; attach it here after capturing it from the deployed site. Do not mark the screenshot evidence complete until that image is attached.


Abstract Algebra Lab
00 Overview
01 Foundations
02 Group Explorer
03 The Lab
04 Algebraic Structure
05 Experiments
06 The Team
07 Research Log
03 / INTERACTIVE CRYPTOGRAPHY LAB
Turn the key.
Trace the proof.
One engine powers the input, the output, the equation and the inverse. Change a parameter and watch the system stay honest.
TRANSFORMATION
LIVE
PLAINTEXT MESSAGE
QWERTYUIOPASDFGHJKL
Shift key
3
Encrypt Message
Decrypt Message
Reset experiment
TZHUWBXLRSDVGIJKMNO
CYBERNETIC CRYPTOGRAPHIC WHEEL / C26
ACTIVE / LIVE
SHIFT KEY
+3
mod 26
KEY 3
DYNAMIC CALCULATION TRACE
Encryption Arithmetic Trace
FORWARD / ENCRYPT
ACTIVE ENCRYPTION RULE
Calculate the ciphertext value by adding the shift and reducing modulo 26.
$$E_3(x) = (x + 3) \bmod 26$$
STEP
PLAIN LETTER
KEY
MATHEMATICAL DERIVATION
CIPHER OUTPUT
01 Q x = 16 shift k = 3 (16 + 3) mod 26 = 19 T y = 19
02 W x = 22 shift k = 3 (22 + 3) mod 26 = 25 Z y = 25
03 E x = 4 shift k = 3 (4 + 3) mod 26 = 7 H y = 7
04 R x = 17 shift k = 3 (17 + 3) mod 26 = 20 U y = 20
05 T x = 19 shift k = 3 (19 + 3) mod 26 = 22 W y = 22
06 Y x = 24 shift k = 3 (24 + 3) mod 26 = 27 mod 26 B y = 1
07 U x = 20 shift k = 3 (20 + 3) mod 26 = 23 X y = 23
08 I x = 8 shift k = 3 (8 + 3) mod 26 = 11 L y = 11
09 O x = 14 shift k = 3 (14 + 3) mod 26 = 17 R y = 17
10 P x = 15 shift k = 3 (15 + 3) mod 26 = 18 S y = 18
11 A x = 0 shift k = 3 (0 + 3) mod 26 = 3 D y = 3
12 S x = 18 shift k = 3 (18 + 3) mod 26 = 21 V y = 21
13 D x = 3 shift k = 3 (3 + 3) mod 26 = 6 G y = 6
14 F x = 5 shift k = 3 (5 + 3) mod 26 = 8 I y = 8
15 G x = 6 shift k = 3 (6 + 3) mod 26 = 9 J y = 9
16 H x = 7 shift k = 3 (7 + 3) mod 26 = 10 K y = 10
17 J x = 9 shift k = 3 (9 + 3) mod 26 = 12 M y = 12
18 K x = 10 shift k = 3 (10 + 3) mod 26 = 13 N y = 13
19 L x = 11 shift k = 3 (11 + 3) mod 26 = 14 O y = 14
19 characters transformed
Cyclic group of order 26
ENGINE TRACE
$E_3(x) = (x + 3) \bmod 26$
INVERSE CHECK
$D^{-1}(E(x)) = x$
verified

Attached file: `ANANYA_14_dynamic-unique-mapping-9plus_2026-10-18.png`